

KW007 Radar Sensor GUI User Guide

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1. Introduction

The KW007 GUI application is a configuration and monitoring tool for the KW007 radar sensor module. This software allows you to:

- Configure sensor detection parameters
- Monitor real-time detection status

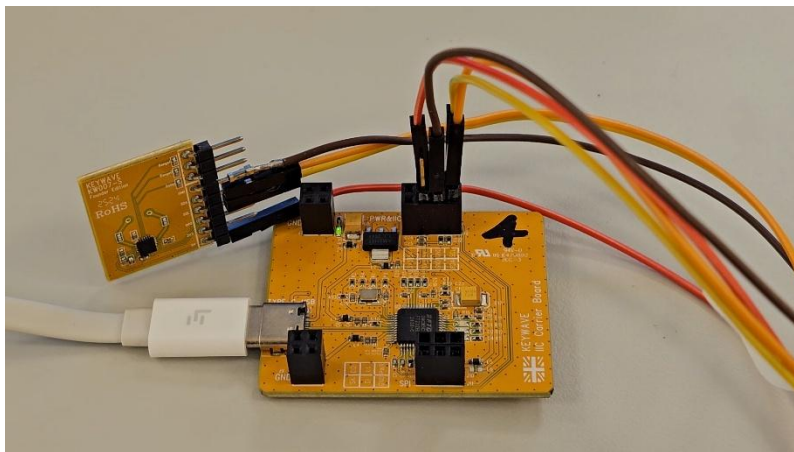
2. System Requirements

- **Operating System:** Windows 10 or later
- **Hardware:** FT232H or CH341 USB-to-I2C adapter
- **Driver:** FTDI D2XX driver (FT232H) or WCH CH341PAR driver (CH341) installed
- **Connection:** KW007 module connected via I2C

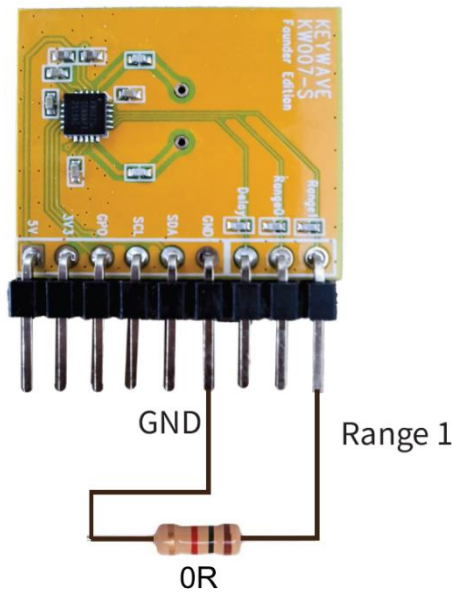
3. Getting Started

3.1 Connecting the Hardware

1. Connect the FTDI USB adapter to your computer
2. Connect the KW007 module to the I2C pins:
 - SDA (Data)
 - SCL (Clock)
 - VCC (Power)
 - GND (Ground)
 - Range1 (Ground)



KEYWAVE



3.2 Hardware Note

When using KW007 in **I²C mode**, the IC pin 19 (Range1) must connect to GND.

3.3 Launching the Application

1. Double-click KW007_DemoUpDn_v2.7.exe to start the application
2. The main window will appear with all controls

4. Main Window Overview

Step1:
Click "Set Initial".

Step2:
Set "Distance Range".

Step3:
Turn on Detection.

FT232H detected
VCO bank (Reg1[5:0]) = 20
Set Initial => OK (verified)
Wrote 32 registers (R0~R31)
R5 (RSSI) = 254
UI updated from chip registers

Adapter locked: FT232H
HW Flag Mode: Started
Detect Period = 128 ms

Approach

Leaving

Two Large circle indicators Show detection status.

Custom Settings

Detection Range

Amplifier 1 Gain 32dB

Amplifier 2 Gain 32dB

Amplifier 3 Gain 8dB

Mag Threshold 48

☐ Enhanced Range Mode

Doppler Filter Corner

High-pass Corner Low

Low-pass Corner High

Sampling Rate 1kHz

GPO Delay 1s

Save

Load

When Distance Range set to (custom), user can manually adjust detail setting.

☒ IIR low pass filter coef: 0.15

Value: 204

RSSI (0-255)

255

192

128

64

0

0 20 40 60 80 100

samples

*USB-to-I2C adapter: FT232H or CH341.

*Distance Range: Range1 is shortest, and Range8 is longest.

5. For “Advanced Experts”

Users can customize settings following the initial setup, or fine-tune them after configuring the distance range.

Detection Range

The detection range depends on two key parameters:

- Amplifier Gain: Increasing the gain extends the range.
- Magnitude Threshold: Lowering the threshold extends the range.

While higher gain or lower thresholds increase range, they also heighten susceptibility to false triggers.

Amplifier Gain Configuration

The total system gain is determined by summing the gain values of the three amplifier stages:

$$Total\ Gain = Amp_1\ Gain + Amp_2\ Gain + Amp_3\ Gain$$

Gain Selection Table

Bit Value	Amp ₁ Gain (dB)	Amp ₂ Gain (dB)	Amp ₃ Gain (dB)
000	8	8	0
001	12	12	4
010	16	16	8
011	19	19	12
100	21	21	-
101	24	24	-
110	28	28	-
111	32	32	-

For optimal Signal-to-Noise Ratio (SNR), it is recommended to apply higher gain settings in the earlier stages ($Amp_1 > Amp_2 > Amp_3$).

Magnitude Threshold Configuration

A target is detected only when the signal magnitude exceeds this threshold. For optimal results, follow the recommended setup sequence below:

1. Measure Baseline RSSI

The “Received Signal Strength Indicator” (RSSI) reflects the average environment background energy when no target is present. Click “Enable RSSI” to view the live noise floor in the bottom-right panel.

2. Set “Mag Threshold” Above Noise Floor

3. Refer to live RSSI to tune “Mag Threshold” for optimal sensitivity

Note:

- Signal Saturation:

Improper gain causes signal saturation (clipping at 0 or 255), leading to detection failure.

- False Triggers:

If “Mag Threshold” is too close to baseline RSSI, environmental noise will trigger false detections.

- “Enhanced Range Mode” uses optimized parameters to boost total detection distance.

Doppler Filter Corner

This section configures the corner frequencies for the Doppler filters:

Filter Type	Function	Trade-off / Impact
Low-Pass Filter (LPF)	Attenuate high-speed clutter and high-frequency noise	Setting too low may attenuate fast-walking targets
High-Pass Filter (HPF)	Reject slow-motion background clutter	Setting too high may miss subtle, slow human movements

Sampling Rate

Key considerations for sampling rate adjustment:

- Filter Scaling:

The Doppler filter corner frequencies scale proportionally with the sampling rate.

- **Noise Reduction:**

A higher sampling rate mitigates noise aliasing, effectively lowering the in-band background noise floor.

GPO Delay

Specifies the hold time for GPO.

- Idle(GPO low): Default state when no motion is detected.
- Trigger(GPO active high): GPO is triggered instantly upon motion detection.
- Release: GPO is held active after motion stops, returning to Idle after delay expires.

Detect Period

I2C polling interval, adjustable from 10 to 1000 ms. The default of 128 ms matches the sensor's update rate; polling faster returns unchanged values.

Save / Load

Save writes the current Custom Adjust Settings and the Detect Period to an .ini file. Load reads one back, applies it to the sensor immediately, and reports the result in the message area. Use this to carry a tuned configuration between installations.

5.1 Monitoring

Approach and Leaving indicators:

Indicator	Color	Status
Upper	Green	Target Approaching
Lower	Blue	Target Leaving
Both	Grey	Idle (No Detection)

Signal timer:

The timer accumulates only while either indicator is lit, so it shows the total time the sensor has reported motion rather than the elapsed run time. Reset returns it to zero.

RSSI chart:

The chart plots reflection strength over the most recent 100 samples, oldest on the left. The vertical scale is 0 to 255; the horizontal scale is in samples, 20 per division. Value: is the latest reading.

IIR low pass filter:

Signal Smoothing: Lower coefficient values apply stronger smoothing, while higher values track raw data closer.

6. Troubleshooting

Message	Cause / Action
"FT232H not found" / "CH341 not found"	The adapter is not plugged in, or Windows has not enumerated it. Reconnect it and try again.
"... busy (used by another app?)"	Another program already has the adapter open. Close it and try again.
"driver missing (ftd2xx.dll)" / "driver missing (CH341DLL.dll)"	Install the FTDI D2XX driver for the FT232H, or the WCH CH341PAR driver for the CH341.
"... disconnected", followed by "stopped, controls unlocked"	The adapter was removed while running. Detection stops and the controls are released.
"Set Initial => FAIL", or a line such as "REG27 FAIL: w=0x30 r=0x00"	The value written did not read back. Check the wiring and the supply, then retry.
Settings appear to be accepted but the sensor never reports motion	Check that pin 19 (Range1) is tied to GND. Without it the sensor runs in hardware mode and ignores the configuration.